

A COUNTEREXAMPLE TO DAVILA'S CLAW-FREE ZOMBIE-DAMAGE CONJECTURE

ABSTRACT. Davila conjectured that every connected claw-free graph G satisfies $\text{zdmg}(G) \leq 4 \text{dmg}(G)$. We construct a connected claw-free graph on nine vertices with $(\text{dmg}, \text{zdmg}) = (2, 9)$, disproving Conjecture 22 of [2]. A ten-vertex extension has values $(2, 10)$ and yields $c_3 \geq 5$ in Davila's notation. An exact exhaustive computation shows that order nine is minimum and that there are exactly 151 isomorphism classes of minimum-order counterexamples.

1. THE COUNTEREXAMPLE

All graphs are finite, simple, and connected. A graph is claw-free if it has no induced $K_{1,3}$. Write $N(v)$ and $N[v]$ for the open and closed neighborhoods of a vertex v . In the one-cop damage game, the cop and robber choose initial vertices in that order, the cop moves first, and both players may move to a neighbor or pass. Capture occurs when the cop moves onto the robber. A vertex occupied by the robber is *damaged* if the robber survives the next cop move and then completes a move, possibly a pass, from that vertex. The damage number $\text{dmg}(G)$ is the number of distinct damaged vertices under optimal play. In the zombie damage game, the pursuer must move at each turn to a neighbor one step closer to the survivor, while the survivor may move or pass; the corresponding value is $\text{zdmg}(G)$. See [1, 2].

Davila's Conjecture 22 states that

$$(1) \quad \text{zdmg}(G) \leq 4 \text{dmg}(G)$$

for every connected claw-free graph G [2, Conjecture 22].

We use the following local form of Davila's trace argument [2, Lemma 12].

Lemma 1 (Local trace lemma). *Let*

$$a_{-2}, a_{-1}, a_0, a_1, \dots, a_m$$

be a walk such that $a_{i-1}a_i a_{i+1}$ is the unique geodesic between a_{i-1} and a_{i+1} for every $-1 \leq i \leq m-1$. If the zombie starts at a_{-2} and the survivor starts at a_0 , then the survivor can damage every vertex occurring among $a_0, \dots, a_m$ by following the walk and passing once at a_m .

Proof. Suppose immediately before a zombie move that the zombie is at a_{i-2} and the survivor is at a_i . The unique-geodesic hypothesis forces the zombie

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to move to a_{i-1} . If $i < m$, the survivor moves to a_{i+1} and damages a_i . Starting with $i = 0$ and iterating gives the claim; when $i = m$, the survivor passes after the zombie moves to a_{m-1} , thereby damaging a_m . $\square$

For $t \geq 1$, define G_t from the cycle

$$v_0 v_1 v_2 v_3 v_4 v_5 v_0$$

by adding a vertex x adjacent to v_0 and v_1 and a path

$$x u_1 u_2 \cdots u_t.$$

Thus $|V(G_t)| = 7 + t$.

Theorem 2. *For $t \in \{2, 3\}$, the graph G_t is connected and claw-free, and*

$$\text{dmg}(G_t) = 2, \quad \text{zdmg}(G_t) = 7 + t.$$

Proof. The only vertices of degree three are v_0, v_1, x . Their neighborhoods contain the edges $v_1 x$, $v_0 x$, and $v_0 v_1$, respectively. Hence G_t is claw-free.

Every vertex is within distance three of x , and $d(x, v_3) = 3$, so $\text{ecc}(x) = 3$. Also $d(u_t, v_3) = t + 3 \geq 5$. Therefore

$$3 = \text{ecc}(x) \geq \text{rad}(G_t) \geq \left\lceil \frac{\text{diam}(G_t)}{2} \right\rceil \geq 3,$$

so $\text{rad}(G_t) = 3$. The bound $\text{dmg}(G) \geq \text{rad}(G) - 1$ gives $\text{dmg}(G_t) \geq 2$ [3, Theorem 1.3].

For the reverse inequality, the cop starts at x . A robber starting at v_0, v_1 , or u_1 is captured on the first cop move. If it starts at u_j with $j \geq 2$, the cop moves to u_1 and remains there, confining the robber to $\{u_2, \dots, u_t\}$, a set of at most two vertices.

For starts on the cycle, use the following two-vertex traps. When the robber occupies an endpoint of the listed edge, the cop occupies the corresponding guard vertex.

robber edge	guard at first endpoint	guard at second endpoint
$v_2 v_3$	v_0	v_5
$v_3 v_4$	v_1	v_0
$v_4 v_5$	v_2	v_1

In each row the two guards are adjacent, and every move from the listed edge through an unlisted neighbor lands next to the current guard. Thus the cop can change guards when the robber crosses the edge and captures on the next cop move if the robber leaves it. A start at v_2 or v_5 is placed in the first or third trap by the move $x \rightarrow v_0$ or $x \rightarrow v_1$, respectively. For a start at v_3 or v_4 , the cop passes at x until the robber first makes a non-pass move and then moves to v_0 or v_1 , respectively. If the robber never makes such a move, it damages only its initial vertex. Thus at most two vertices are damaged, proving $\text{dmg}(G_t) \leq 2$.

It remains to prove full zombie damage. Write

$$U_t = (x, u_1, \dots, u_t),$$

and use U_t as a terminal word in the displays below. For the indicated initial zombie vertices, consider the following walks; in each row the survivor starts at the third displayed vertex:

$$\begin{aligned} T_{v_0} &= (v_0, v_5, v_4, v_3, v_2, v_1, v_0, v_5, v_4, v_3, v_2, v_1, U_t), \\ T_{v_2} &= (v_2, v_1, v_0, v_5, v_4, v_3, v_2, v_1, U_t), \\ T_{v_3} &= (v_3, v_2, v_1, v_0, v_5, v_4, v_3, v_2, v_1, U_t), \\ T_x &= (x, v_1, v_2, v_3, v_4, v_5, v_0, v_1, v_2, v_3, v_4, v_5, v_0, U_t), \\ T_{u_1} &= (u_1, x, v_0, v_5, v_4, v_3, v_2, v_1, U_t). \end{aligned}$$

The survivor portion of every walk visits all vertices of G_t . Every length-two segment lying on the 6-cycle or the pendant path is the unique geodesic between its endpoints. The remaining consecutive triples are, up to reversal,

$$(v_2, v_1, x), \quad (v_5, v_0, x), \quad (v_1, x, u_1), \quad (v_0, x, u_1).$$

For each of these four triples, the endpoints are nonadjacent and the displayed middle vertex is their unique common neighbor. Hence every consecutive triple satisfies the hypothesis of Lemma 1.

The automorphism interchanging

$$v_0 \leftrightarrow v_1, \quad v_2 \leftrightarrow v_5, \quad v_3 \leftrightarrow v_4$$

and fixing $x, u_1, \dots, u_t$ covers initial zombie vertices v_1, v_5, v_4 . Finally, if the zombie starts at u_j with $j \geq 2$, the survivor starts at v_0 and passes while the zombie is forced toward u_1 . After the zombie first reaches u_1 , the survivor passes once more; the next zombie turn then begins in the initial position of T_{u_1} . Thus the survivor damages every vertex against every initial zombie position, and $\text{zdmg}(G_t) = |V(G_t)| = 7 + t$. $\square$

Davila defines

$$c_3 = \sup \left\{ \frac{\text{zdmg}(G)}{\text{dmg}(G)} : G \text{ is connected and claw-free, } \text{dmg}(G) > 0 \right\}.$$

Corollary 3. *The graph G_2 disproves Davila's Conjecture 22, since*

$$\text{zdmg}(G_2) = 9 > 8 = 4 \text{dmg}(G_2).$$

Moreover,

$$\frac{\text{zdmg}(G_3)}{\text{dmg}(G_3)} = 5,$$

so $c_3 \geq 5$.

2. MINIMUM ORDER

Proposition 4. *No connected claw-free graph of order at most eight violates (1). At order nine, exactly 151 isomorphism classes violate (1), all with $(\text{dmg}, \text{zdmg}) = (2, 9)$. Hence G_2 has minimum possible order.*

Computational proof. Let $P \in \{C, Z\}$ denote an ordinary cop or a zombie, write $\text{val}_C(G) = \text{dmg}(G)$ and $\text{val}_Z(G) = \text{zdmg}(G)$, and put

$$M_C(p, s) = N[p], \quad M_Z(p, s) = \{p' \in N(p) : d(p', s) = d(p, s) - 1\}.$$

Fix an integer k . A state (D, p, s) consists of the damaged set D and distinct current pursuer and survivor positions p, s , immediately before a pursuer move. Let W_k^P be the least set of states that contains every state with $|D| \geq k$ and satisfies the following closure rule: add (D, p, s) if, for every $p' \in M_P(p, s)$, one has $p' \neq s$ and either $|D \cup \{s\}| \geq k$ or there exists

$$s' \in N[s] \setminus \{p'\}$$

with $(D \cup \{s\}, p', s') \in W_k^P$.

A state added after r iterations is one from which the survivor can force k damaged vertices within at most r further rounds. Conversely, if a state is outside the stabilized set, failure of the closure rule gives the pursuer a move that either captures immediately or leaves every legal survivor reply outside the set while the damaged total remains below k . Repeating such choices prevents the threshold forever. Therefore

$$\text{val}_P(G) \geq k \iff \forall p \in V(G) \exists s \in V(G) \setminus \{p\} : (\emptyset, p, s) \in W_k^P.$$

We evaluated this recurrence exactly on one representative of each connected isomorphism class through order nine, generated by `geng -c n` from NAUTY [4], and retained the claw-free graphs by checking every triple in each open neighborhood. The results were

n	connected classes	claw-free classes	counterexample classes
1	1	1	0
2	1	1	0
3	2	2	0
4	6	5	0
5	21	14	0
6	112	50	0
7	853	191	0
8	11117	881	0
9	261080	4494	151

All 151 counterexamples had values $(2, 9)$. Using the vertex order

$$v_0, \dots, v_5, x, u_1, \dots, u_t,$$

the graph6 strings of G_2 and G_3 are, respectively,

$$\text{HhEM?C@} \quad \text{and} \quad \text{IhEM?C@?G}.$$

□

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